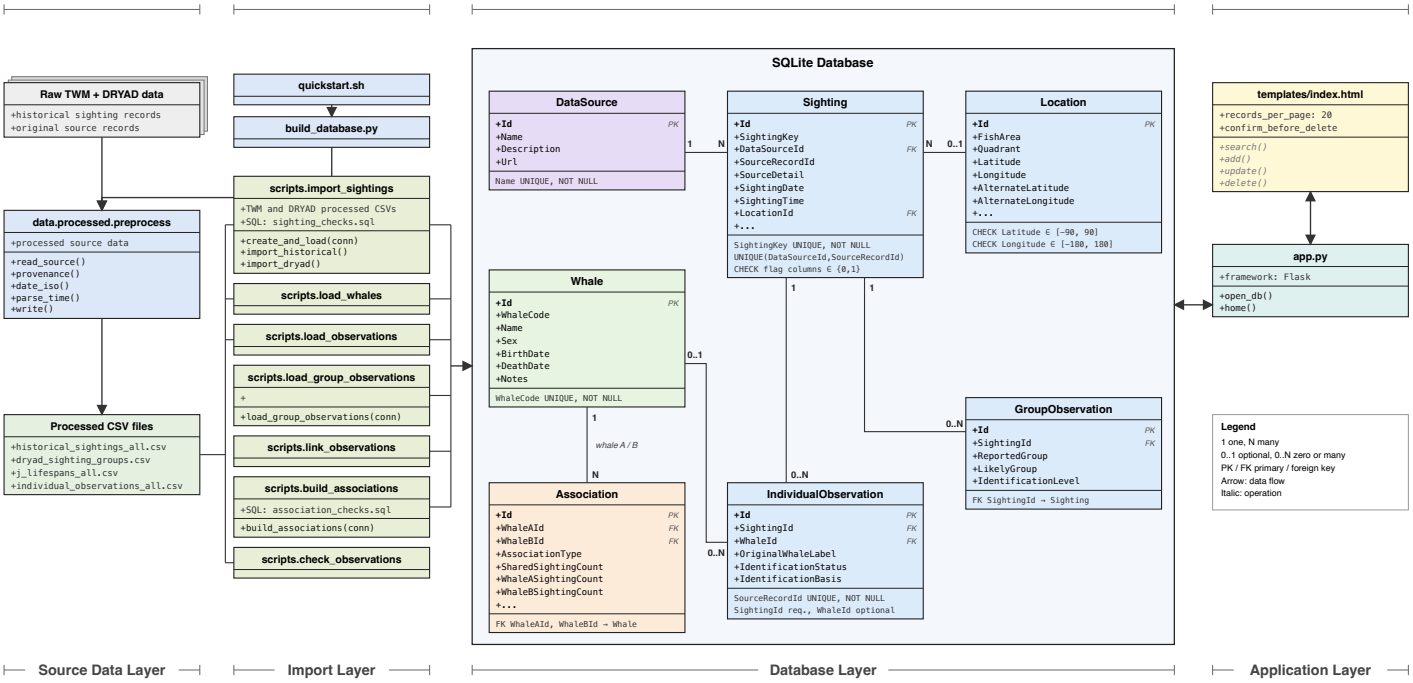

Raw TWM + DRYAD data
+historical sighting records
+original source records
data.processed.preprocess
+processed source data
+read_source()
+provenance()
+date_iso()
+parse_time()
+write()
Processed CSV files
+historical_sightings_all.csv
+dryad_sighting_groups.csv
+j_lifespans_all.csv
+individual_observations_all.csv
quickstart.sh
build_database.py
scripts.import_sightings
+TWM and DRYAD processed CSVs
+SQL: sighting_checks.sql
+create_and_load(conn)
+import_historical()
+import_dryad()
scripts.load_whales
scripts.load_observations
scripts.load_group_observations
+
+load_group_observations(conn)
scripts.link_observations
scripts.build_associations
+SQL: association_checks.sql
+build_associations(conn)
scripts.check_observations
SQLite Database
DataSource
+Id PK
+Name
+Description
+Url
Name UNIQUE, NOT NULL
Sighting
+Id PK
+SightingKey
+DataSourceId FK
+SourceRecordId
+SourceDetail
+SightingDate
+SightingTime
+LocationId FK
+...
SightingKey UNIQUE, NOT NULL
UNIQUE(DataSourceId,SourceRecordId)
CHECK flag columns ∈ {0,1}
Location
+Id PK
+FishArea
+Quadrant
+Latitude
+Longitude
+AlternateLatitude
+AlternateLongitude
+...
CHECK Latitude ∈ [−90, 90]
CHECK Longitude ∈ [−180, 180]
Whale
+Id PK
+WhaleCode
+Name
+Sex
+BirthDate
+DeathDate
+Notes
WhaleCode UNIQUE, NOT NULL
whale A / B
Association
+Id PK
+WhaleAId FK
+WhaleBId FK
+AssociationType
+SharedSightingCount
+WhaleASightingCount
+WhaleBSightingCount
+...
FK WhaleAId, WhaleBId → Whale
IndividualObservation
+Id PK
+SightingId FK
+WhaleId FK
+OriginalWhaleLabel
+IdentificationStatus
+IdentificationBasis
SourceRecordId UNIQUE, NOT NULL
SightingId req., WhaleId optional
GroupObservation
+Id PK
+SightingId FK
+ReportedGroup
+LikelyGroup
+IdentificationLevel
FK SightingId → Sighting
templates/index.html
+records_per_page: 20
+confirm_before_delete
+search()
+add()
+update()
+delete()
app.py
+framework: Flask
+open_db()
+home()
Legend
1 one, N many
0..1 optional, 0..N zero or many
PK / FK primary / foreign key
Arrow: data flow
Italic: operation
Source Data Layer
Import Layer
Database Layer
Application Layer